CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Complementary events

1. A die is rolled once. Find the probability that the number shown is not 3. (a) $\frac{1}{6}$ (b) $\frac{3}{6}$ (c) $\frac{5}{6}$ (d) 1 CBSE 2026, 1 mark

SOLUTION

$P(3) = \frac{1}{6}$, so $P(\text{not } 3) = 1 - \frac{1}{6} = \frac{5}{6}$

Answer: (c)

2. A player wins a game with probability 0.79. What is the probability that the player loses it? (a) 1.79 (b) 0.31 (c) 0.21% (d) 0.21 CBSE 2024, 1 mark

SOLUTION

$P(\text{losing}) = 1 - P(\text{winning}) = 1 - 0.79 = 0.21$

Answer: (d)

3. An event happens with probability p and does not happen with probability q. How are p and q related? (a) $p + q = 1$ (b) $p = 1, q = 1$ (c) $p = q - 1$ (d) $p + q + 1 = 0$ CBSE 2023, 1 mark

SOLUTION

An event and its complement cover all the outcomes, so their probabilities add up to 1.

Answer: (a)

4. In a group of 20 people, 5 cannot swim. One person is chosen at random. Find the probability that the person can swim. (a) $\frac{3}{4}$ (b) $\frac{1}{3}$ (c) 1 (d) $\frac{1}{4}$ CBSE 2023, 1 mark

SOLUTION

$20 - 5 = 15$ people can swim. Probability $= \frac{15}{20} = \frac{3}{4}$

Answer: (a)

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5. In a game a coin is tossed 3 times. A player wins if all three tosses give the same result. Find the probability of losing. CBSE 2019, 2 marks

SOLUTION

There are $2 \times 2 \times 2 = 8$ equally likely outcomes. Only HHH and TTT are wins.

$P(\text{win}) = \frac{2}{8} = \frac{1}{4}$, so $P(\text{lose}) = 1 - \frac{1}{4}$

Answer: $\frac{3}{4}$

6. An integer is chosen at random from those lying between 1 and 100. Find the probability that it is (i) divisible by 8 (ii) not divisible by 8. CBSE 2018, 2 marks

SOLUTION

The integers between 1 and 100 are $2, 3, \ldots, 99$: 98 numbers.

(i) Multiples of 8 among them: $8, 16, \ldots, 96$, which is 12 numbers. Probability $= \frac{12}{98} = \frac{6}{49}$.

(ii) $1 - \frac{6}{49} = \frac{43}{49}$

Answer: (i) $\frac{6}{49}$ **(ii)** $\frac{43}{49}$

2 Theoretical probability

7. Assertion (A): The probability that a leap year has 53 Mondays is $\frac{2}{7}$. Reason (R): The probability that a non-leap year has 53 Mondays is $\frac{5}{7}$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2026, 1 mark

SOLUTION

A leap year has 366 days $=$ 52 weeks $+$ 2 extra days. The extra days can be (Sun, Mon), (Mon, Tue), ..., (Sat, Sun): 7 equally likely pairs, and 2 of them contain a Monday. So A is true.

A non-leap year has 365 days $=$ 52 weeks $+$ 1 extra day, which is a Monday with probability $\frac{1}{7}$, not $\frac{5}{7}$. So R is false.

Answer: (c)

8. Assertion (A): When one number is picked at random from the numbers 1 to 20, the probability that it is a number from 1 to 20 is 1. Reason (R): An event E with $P(E) = 1$ is a sure event. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2025, 1 mark

SOLUTION

Every possible outcome is a number from 1 to 20, so the event is certain to happen and its probability is $\frac{20}{20} = 1$. A is true.

R is true, and it is exactly the reason for A.

Answer: (a)

9. The even numbers are removed from the list 1, 4, 7, 9, 16, 21, 25, and one of the remaining numbers is picked at random. Find the probability that it is prime. (a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{7}$ (d) $\frac{2}{7}$ CBSE 2024, 1 mark

SOLUTION

Removing the even numbers 4 and 16 leaves 1, 7, 9, 21, 25: five numbers.

The only prime among them is 7 (1 is not prime; $9 = 3^2$, $21 = 3 \times 7$, $25 = 5^2$).

Probability $= \frac{1}{5}$

Answer: (b)

10. 6000 lottery tickets are sold. A girl works out that her probability of winning the first prize is 0.08. How many tickets did she buy? (a) 40 (b) 240 (c) 480 (d) 750 CBSE 2023, 1 mark

SOLUTION

$\frac{\text{her tickets}}{6000} = 0.08$, so her tickets $= 0.08 \times 6000 = 480$.

Answer: (c)

11. A bag holds 4 red, 3 blue and 2 yellow balls. One ball is drawn at random. Find the probability that it is (i) red (ii) yellow. CBSE 2023, 2 marks

SOLUTION

Total balls $= 4 + 3 + 2 = 9$

Answer: (i) $\frac{4}{9}$ **(ii)** $\frac{2}{9}$

12. A die is thrown once. Find the probability of getting (i) a prime number (ii) a number lying between 2 and 6. CBSE 2019, 2 marks

SOLUTION

(i) The primes on a die are 2, 3 and 5: probability $= \frac{3}{6} = \frac{1}{2}$.

(ii) The numbers between 2 and 6 are 3, 4 and 5: probability $= \frac{3}{6} = \frac{1}{2}$.

Answer: (i) $\frac{1}{2}$ **(ii)** $\frac{1}{2}$

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13. Three different coins are tossed together. Find the probability of getting (i) exactly two heads (ii) at least two heads (iii) at least two tails. CBSE 2016, 3 marks

SOLUTION

The 8 equally likely outcomes are HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

(i) Exactly two heads: HHT, HTH, THH. Probability $= \frac{3}{8}$.

(ii) At least two heads: HHT, HTH, THH, HHH. Probability $= \frac{4}{8} = \frac{1}{2}$.

(iii) At least two tails: HTT, THT, TTH, TTT. Probability $= \frac{4}{8} = \frac{1}{2}$.

Answer: (i) $\frac{3}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{2}$

14. A number x is chosen at random from 1, 2, 3, 4, and a number y is chosen at random from 1, 4, 9, 16. Find the probability that the product xy is less than 16. CBSE 2016, 4 marks

SOLUTION

There are $4 \times 4 = 16$ equally likely pairs (x, y).

$x = 1$: $y = 1, 4, 9$ (3 pairs). $x = 2$: $y = 1, 4$ (2 pairs). $x = 3$: $y = 1, 4$ (2 pairs). $x = 4$: $y = 1$ (1 pair, since $4 \times 4 = 16$ is not less than 16).

Favourable pairs $= 3 + 2 + 2 + 1 = 8$

Answer: $\frac{8}{16} = \frac{1}{2}$

3 Two dice

15. Two dice of different colours are thrown together. List all the possible outcomes. Find the probability that (i) both dice show the same number (ii) the dice show different numbers. CBSE 2026, 3 marks

SOLUTION

Write each outcome as (number on the first die, number on the second die). The outcomes are $(1,1), (1,2), \ldots, (1,6), (2,1), \ldots, (2,6)$, and so on up to $(6,1), \ldots, (6,6)$: $6 \times 6 = 36$ in all.

(i) Same number: $(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)$, which is 6 outcomes. Probability $= \frac{6}{36} = \frac{1}{6}$.

(ii) Different numbers: $36 - 6 = 30$ outcomes. Probability $= \frac{30}{36} = \frac{5}{6}$.

Answer: (i) $\frac{1}{6}$ (ii) $\frac{5}{6}$

16. Two dice are thrown together. Find the probability that the two numbers differ by 2. CBSE 2025, 3 marks

SOLUTION

There are $6 \times 6 = 36$ equally likely outcomes.

Pairs that differ by 2: $(1,3), (3,1), (2,4), (4,2), (3,5), (5,3), (4,6), (6,4)$, which is 8 outcomes.

Probability $= \frac{8}{36}$

Answer: $\frac{2}{9}$

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17. Two dice are thrown together. Find the probability that the two numbers add up to 2, 3 or 5. (a) $\frac{7}{36}$ (b) $\frac{11}{36}$ (c) $\frac{5}{36}$ (d) $\frac{4}{9}$ CBSE 2024, 1 mark

SOLUTION

Sum 2: $(1,1)$. Sum 3: $(1,2),(2,1)$. Sum 5: $(1,4),(2,3),(3,2),(4,1)$.

Favourable outcomes $= 1+2+4 = 7$ out of 36.

Answer: (a)

18. Two different dice are thrown together. Find the probability of getting (i) a doublet (ii) a sum of 10. CBSE 2018, 2 marks

SOLUTION

There are 36 equally likely outcomes.

(i) Doublets: $(1,1),(2,2),\ldots,(6,6)$, which is 6 outcomes. Probability $= \frac{6}{36} = \frac{1}{6}$.

(ii) Sum 10: $(4,6),(5,5),(6,4)$, which is 3 outcomes. Probability $= \frac{3}{36} = \frac{1}{12}$.

Answer: (i) $\frac{1}{6}$ (ii) $\frac{1}{12}$

4 Playing cards

19. One card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is a red face card. CBSE 2025, 1 mark

SOLUTION

The face cards are the jack, queen and king. The two red suits (hearts and diamonds) have $2 \times 3 = 6$ of them.

Probability $= \frac{6}{52}$

Answer: $\frac{3}{26}$

20. A black card is missing from a pack of 52 playing cards. One card is drawn at random from the rest. Find the probability that it is the queen of hearts. CBSE 2024, 2 marks

SOLUTION

One black card is lost, so 51 cards remain.

The queen of hearts is a red card, so it is still in the pack.

Answer: $\frac{1}{51}$

21. A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is neither a red card nor a queen. CBSE 2016, 1 mark

SOLUTION

Remove the 26 red cards. Of the 26 black cards, 2 are queens. That leaves $26 - 2 = 24$ cards that are neither red nor a queen.

Answer: $\frac{24}{52} = \frac{6}{13}$

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